

The Program in Action

When you run *diceroller.py*, the output will look like this:

```
Dice Roller, by Al Sweigart al@inventwithpython.com
--snip--
> 3d6
7 (3, 2, 2)
> 1d10+2
9 (7, +2)
> 2d38-1
32 (20, 13, -1)
> 100d6
364 (3, 3, 2, 4, 2, 1, 4, 2, 4, 6, 4, 5, 4, 3, 3, 3, 2, 5, 1, 5, 6, 6, 6, 4,
5, 5, 1, 5, 2, 2, 2, 5, 1, 1, 2, 1, 4, 5, 6, 2, 4, 3, 4, 3, 5, 2, 2, 1, 1, 5,
1, 3, 6, 6, 6, 6, 5, 2, 6, 5, 4, 4, 5, 1, 6, 6, 6, 4, 2, 6, 2, 6, 2, 2, 4, 3,
6, 4, 6, 4, 2, 4, 3, 3, 1, 6, 3, 3, 4, 4, 5, 5, 5, 6, 2, 3, 6, 1, 1, 1)
--snip--
```

How It Works

Most of the code in this program is dedicated to ensuring that the input the user entered is properly formatted. The actual random dice rolls themselves are simple calls to `random.randint()`. This function has no bias: each integer in the range passed to it is equally likely to be returned. This makes `random.randint()` ideal for simulating dice rolls.

```
1. """Dice Roller, by Al Sweigart al@inventwithpython.com
2. Simulates dice rolls using the Dungeons & Dragons dice roll notation.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: short, simulation"""
5.
6. import random, sys
7.
8. print('''Dice Roller, by Al Sweigart al@inventwithpython.com
9.
10. Enter what kind and how many dice to roll. The format is the number of
11. dice, followed by "d", followed by the number of sides the dice have.
12. You can also add a plus or minus adjustment.
13.
14. Examples:
15.   3d6 rolls three 6-sided dice
16.   1d10+2 rolls one 10-sided die, and adds 2
17.   2d38-1 rolls two 38-sided die, and subtracts 1
18.   QUIT quits the program
19. ''')
20.
21. while True: # Main program loop:
22.     try:
23.         diceStr = input('> ') # The prompt to enter the dice string.
24.         if diceStr.upper() == 'QUIT':
25.             print('Thanks for playing!')
26.             sys.exit()
```